

EMPLOYEE DATA ANALYSIS PYTHON PROJECT REPORT

1. Introduction

This project focuses on the analysis of an employee dataset using Python. The purpose of the analysis is to explore employee characteristics, salary patterns, education levels, industries, remote-work arrangements, skills, certifications, and years of experience.

The analysis was carried out using Pandas, NumPy, Plotly, and Scikit-learn. Exploratory Data Analysis (EDA) was performed to identify patterns and relationships within the dataset. Calculated columns, filtering and sorting operations, correlation analysis, visualizations, descriptive statistics, and a Logistic Regression model were also developed.

2. Dataset Description

The dataset contains 1,999 employee records and 10 columns.

It contains the following column

- Job title
- Experience years
- Education level
- Skills count
- Industry
- Company size
- Location
- Remote work
- Certifications
- Salary

The dataset was examined for missing values and duplicate records.

It was clean and did not require removal of missing or duplicate records.

3. Exploratory Data Analysis

Exploratory Data Analysis was performed to understand the structure, patterns and characteristics of the dataset.

The analysis examined:

- Data shape
- Data types
- Missing values
- Duplicate records
- Unique values
- Salary distribution
- Job-title distribution
- Salary by education
- Salary by industry
- Remote-work distribution
- Relationship between experience and salary
- Plotly visualizations.

4. Descriptive Statistics

Descriptive statistics were calculated for the numerical variables.

- The average employee has approximately 9.88 years of experience, while the average number of skills is approximately 9.95.
- The average number of certifications is approximately 2.47.
- The average salary is approximately 145,606, while the median salary is approximately 144,185.
- The salary values range from 41,404 to 292,855, showing considerable variation in employee salaries.
- The standard deviation of approximately 36,264 indicates that salaries vary substantially around the average.

5. Calculated Columns

Three calculated columns were created.

- Salary Category

Employees were classified into two categories:

High: salary $\geq$ 150,000

Low: salary < 150,000

This column was later used as the target variable for the Logistic Regression model.

- Experience Category

Employees were divided into:

Junior: 0–5 years

Mid-Level: 6–10 years

Senior: more than 10 years

This makes it easier to analyze employees according to their experience level.

- Salary per Experience

Salary per experience column was created by dividing salary by years of experience.

For employees with zero years of experience, the calculation was handled separately to avoid division by zero.

6. Sorting and Filtering Operations

Five sorting/filtering operations were performed on the dataset.

These included:

- Sorting employees by salary from highest to lowest.
- Sorting employees by experience from lowest to highest.
- Filtering employees with a high salary category.
- Filtering employees with 10 or more years of experience.
- Filtering employees whose remote-work status is Yes.

These operations demonstrated how specific groups of employees can be extracted from the dataset.

7. Index-Based Filtering

Seven different index-based filtering operations were performed using Pandas .iloc.

The operations demonstrated how rows and columns can be selected based on their numerical positions rather than their column names.

The following was performed:

- Selecting the first row.
- Selecting the first five rows.
- Selecting specific rows.
- Selecting a range of rows.
- Selecting specific rows and columns.
- Selecting specific row and column ranges.
- Selecting last rows and columns.

This demonstrated practical use of index-based data selection in Pandas.

8. Correlation Analysis

Correlation analysis was performed on this numerical variable:

- Experience years
- Skills count
- Certifications
- Salary

The correlation between experience and salary was approximately:
0.4109

The correlation between skills and salary was approximately:
0.1346

The correlation between certifications and salary was approximately:
0.1284

Interpretation

- The strongest relationship observed was between experience and salary.
- The correlation of approximately 0.41 indicates a moderate positive linear relationship. This means that employees with more years of experience generally tend to have higher salaries, although experience alone does not determine salary.
- Skills and certifications had weaker positive relationships with salary.

A labelled correlation heatmap was created using Plotly.

9. Data Visualizations

A dashboard containing six visuals was created using Plotly.

The dashboard provides a visual summary of the employee dataset and allows the major findings to be viewed together rather than as separate charts.

These visualizations were created:

- Employee by Job Title using bar chart.
- Salary Distribution using histogram.
- Average Salary by Education using bar chart.
- Average Salary by Industry using bar chart.
- Employees by Remote Work using bar chart.
- Experience vs Salary using scatter graph.

These six visuals were combined into an Employee Data Analysis Dashboard.

The dashboard makes it easier for decision maker to quickly understand the important patterns in the employee data and communicate findings from the dataset.

Key observations

- Job Title: Data Scientist had the highest number of employees in the dataset, while Data Analyst had the lowest among the listed job titles.
- Salary Distribution: Most salaries are concentrated around the middle range, with fewer employees receiving very high salaries.
- Education: The average salary generally differs according to education level. PhD employees had the highest average salary, while High School employees had the lowest average salary.
- Industry: Average salaries varied across industries, with some industries having slightly higher average salaries than others.
- Remote Work: Hybrid work had the largest number of employees, followed by employees who did not work remotely and then those who worked remotely.
- Experience vs Salary: The scatter plot shows a general upward tendency between experience and salary.

10. Logistic Regression

A Logistic Regression model was developed to predict whether an employee belongs to the High or Low Salary category.

The target variable was:

- salary_category

The predictors used were:

- experience_years
- skills_count
- certifications

The dataset was divided into:

80% training data

20% testing data

The model was trained using the training dataset and then used to make predictions on the testing dataset.

11. Logistic Regression Results

The model achieved an accuracy of:

68%

This means that the model correctly predicted the salary category (High or Low) 68 out of every 100 employees in the test data.

The classification report produced the following results:

Class	Precision	recall	F1-score	Support
High	0.66	0.60	0.63	182
Low	0.69	0.75	0.72	218
Accuracy			0.68	400
Macro avg	0.68s	0.67	0.67	400
Weighted avg	0.68	0.68	0.68	400

Interpretation

- The model performed better at identifying Low Salary employees than High Salary employees.
- For High Salary employees, the recall was 60%, meaning the model identified 60% of employees who actually belonged to the High Salary category.
- For Low Salary employees, the recall was 75%, meaning the model correctly identified 75% of employees who actually belonged to the Low Salary category.
- The overall accuracy of 68% indicates that the model has a reasonable predictive ability, although there is room for improvement.

12. Confusion Matrix

The Confusion matrix, using High and Low as the class order is shown below:

	Predicted High	Predicted Low
Actual High	109	73
Actual Low	55	163

The model correctly classified:

109 High Salary employees

163 Low Salary employees

Therefore:

$109 + 163 = 272$ correct predictions

Out of 400 test records:

$272 \div 400 = 68\%$

This confirms the reported model accuracy.

13. What is Linear Regression?

Linear Regression is a supervised machine learning technique used to predict a continuous numerical value.

It attempts to find a relationship between one or more independent variables and a numerical dependent variable.

Real-world use case

A company could use Linear Regression to estimate the expected salary of a new employee based on their experience, education, skills, and certifications.

14. What is Logistic Regression?

Logistic Regression is a supervised machine learning algorithm used mainly for classification problems.

Unlike Linear Regression, which predicts a continuous number, Logistic Regression predicts the probability that an observation belongs to a particular category.

Real-world use case

A company could use Logistic Regression to predict whether a job applicant is likely to be classified as a high-salary employee based on their experience, skills, and certifications. Another common use is predicting whether a customer will leave or stay with a company.

15. Conclusion

The Employee Data Analysis project successfully demonstrated the use of Python for data analysis.

The dataset contained 1,999 employee records, and the data quality assessment showed no missing values or duplicate records.

- Three calculated columns were created to provide additional insight into salary and experience.
- Sorting and filtering operations were performed using both column-based and index-based methods.
- The analysis showed a moderate positive relationship between experience and salary, with a correlation of approximately 0.41. This suggests that experience is an important factor associated with salary, although other factors also contribute.
- Six Plotly visualizations were combined into an Employee Data Analysis Dashboard to provide a visual summary of the dataset.
- Finally, a Logistic Regression model was developed to classify employees into High and Low salary categories. The model achieved an overall accuracy of 68% and performed better at identifying Low Salary employees than High Salary employees.

Overall, the project demonstrates how data cleaning, exploratory analysis, visualization, statistical analysis, and machine learning can be combined to extract useful insights from employee data.